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DETECTION AND RECORDING OF BLOW OUT PREVENTER RAM OPERATIONS DURING RIG-LESS WELLBORE INTERVENTIONS

Abstract

A rig-less wellbore system includes a blowout preventer (BOP) central body fluidly coupled to a wellbore and a plurality of hydraulic cylinders extending radially from the central body. A plurality of pistons operable to move radially within a respective hydraulic cylinder between an open position wherein flow through the central passageway is permitted and a closed position wherein the pistons extend into the central passageway to prohibit fluid flow through the central passageway. At least one sensor is operable to monitor at least one parameter indicative of activity within a hydraulic circuit operably associated with an individual one of the pistons. A data acquisition system (DAS) is communicatively coupled to the at least one sensor to collect and record data provided by the at least one sensor.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates generally to the operation of blowout preventers (BOPs) in wellbore operations, and more particularly, to detecting and monitoring individual events of a BOP ram operation.

BACKGROUND OF THE DISCLOSURE

[0002] Wellbores may be drilled to recover natural deposits of oil and gas, as well as other desirable materials that are trapped in subterranean geological formations. Because fluids may be trapped in the subterranean geological formations at very high pressures, injury to personnel and damage to equipment may result from unexpected events associated with drilling, workover, rig-less interventions, or production activities. Pressure control equipment, such as a blowout preventer (BOP), may be installed at a surface location to prevent uncontrolled release of wellbore fluids.

[0003] A BOP is a mechanical device used to seal wellbore conduits whenever a dangerous condition is detected. One type of BOP is a ram-type BOP, which includes a pair of opposing pistons with blocks or “rams” that may be forced toward one another to seal a wellbore conduit extending through the BOP. Faces of the rams may be fitted with seals that engage the other ram or an outer surface of the wellbore conduit or cable extending through the BOP.

[0004] Knowledge related to the operation of the BOP may be extremely important to maintaining proper operation of the BOP and anticipating future problems. By monitoring the parameters and events of each BOP operation, a wellbore may be more effectively operated such that safe conditions can be maintained. Furthermore, when an unsafe condition is detected, remedial action may be appropriately initiated, either manually or automatically. Information gathered by monitoring BOP events may be useful in post-job review, training purposes, implementing lessons learned and assisting in incident investigations.

SUMMARY OF THE DISCLOSURE

[0005] Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

[0006] According to an embodiment consistent with the present disclosure, a rig-less wellbore system includes a blowout preventer (BOP) central body defining a central passageway extending along a longitudinal axis through the central body. The central passageway is fluidly coupled to a wellbore. The system further includes a plurality of hydraulic cylinders extending radially from the central body and a plurality of pistons. Each piston is operable to move radially within a respective hydraulic cylinder between an open position wherein flow through the central passageway is permitted and a closed position wherein the pistons extend into the central passageway to prohibit fluid flow through the central passageway. At least one sensor is operable to monitor at least one parameter indicative of activity within a hydraulic circuit operably associated with an individual one of the pistons, and a data acquisition system (DAS) is communicatively coupled to the at least one sensor. The DAS operable to collect and record data provided by the at least one sensor.

[0007] According to another example embodiment consistent with the present disclosure, a method of operating and monitoring a rig-less wellbore system includes (a) passing a conveyance through a BOP including a plurality of pistons, (b) detecting, with at least one sensor, at least one parameter indicative of activity within a hydraulic circuit operably associated with an individual piston of the plurality of pistons, (c) transmitting data collected by the at least one sensor about the at least one parameter through a communication cable to a DAS, and (d) observing, with the DAS, a change in the data over time indicative of activity within the hydraulic circuit.

[0008] Any combinations of the various embodiments and implementations disclosed herein can be used in a further embodiment, consistent with the disclosure. These and other aspects and features can be appreciated from the following description of certain embodiments presented herein in accordance with the disclosure and the accompanying drawings and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a cross-sectional view of a wellbore system for conducting rig-less wellbore interventions with a BOP operably coupled to a data acquisition system (DAS) in accordance with one or more aspects of the present disclosure.

[0010] FIG. 2A is an enlarged schematic view of the BOP of FIG. 1 illustrating a control panel through which the BOP is operably coupled to the DAS.

[0011] FIG. 2B is a schematic cross-sectional view of the BOP illustrating a plurality of hydraulically operable rams therein.

[0012] FIG. 3 is a schematic view of a hydraulic circuit for controlling and monitoring each of the individual rams of the BOP.

[0013] FIG. 4 is graphical view of a pressure detected within a hydraulic circuit of a BOP during a test operation in which the rams of the BOP were closed and opened.

DETAILED DESCRIPTION

[0014] Embodiments of the present disclosure will now be described in detail with reference to the accompanying Figures. Like elements in the various figures may be denoted by like reference numerals for consistency. Further, in the following detailed description of embodiments of the present disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the claimed subject matter. However, it will be apparent to one of ordinary skill in the art that the embodiments disclosed herein may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description. Additionally, it will be apparent to one of ordinary skill in the art that the scale of the elements presented in the accompanying Figures may vary without departing from the scope of the present disclosure.

[0015] Embodiments in accordance with the present disclosure generally relate to systems and methods for monitoring the operation of a BOP employed in rig-less wellbore interventions. One or more sensors may be positioned within a hydraulic circuit for actuating the BOP, and the sensors may monitor parameters indicative of the events associated with individual rams of the BOP. The events may include any activity within the hydraulic circuit including changes in pressure or temperature of a pressurized hydraulic fluid, a flow of the pressurized hydraulic fluid, abnormal sounds and/or position changes of the individual rams of the BOP. The sensors may be operatively coupled to a data acquisition system (DAS) that collects, displays, records data from the sensors, and digitizes readings.

[0016] FIG. 1 is a schematic view of a wellbore system **100** including a BOP **102** in accordance with one or more exemplary embodiments of the disclosure. The wellbore system **100** is partially disposed in a wellbore **106** extending from a surface location “S” and traversing a geologic

formation "G." In the illustrated example, the wellbore **106** is substantially vertical. In other embodiments, aspects of the disclosure may be practiced in a wide variety of vertical, directional, deviated, slanted and/or horizontal portions therein, and may extend along any trajectory through the geologic formation "G." As illustrated in FIG. 1, the wellbore **106** is open hole, but in other embodiments, the wellbore may be at least partially lined with a casing string (not shown) without departing from the scope of the disclosure.

[0017] In the example illustrated embodiment, the wellbore system **100** includes a conveyance **110** extending into the wellbore **106** from the surface location "S." The conveyance **110** may be constructed of a continuous string of flexible tubing (coiled tubing), an electrified cable (wireline) or nonelectric cable (slickline) for the selective placement and retrieval of a downhole tool **114** coupled to a downhole end of the conveyance **110**. The downhole tool **114** may include any number of logging tools for gathering pressure, temperature and flow data, perforating tools for perforating the geologic formation, or other tools operable for other interventions to change or adjust downhole equipment such as valves or pumps.

[0018] The conveyance **110** passes through a wellhead **116** at the surface location "S" to a trailer **118**, which may be used to transport and store various components of the wellbore system **100**. For example, the trailer **118** may carry a reel **120**, around which the conveyance **110** is wound, a controller **122** and a power system **124**. The controller **122** may be operably coupled to the BOP **102**, downhole tool **114** and/or other components of the wellbore system **100**. In some embodiments, the controller **122** may be a computer-based system that may include a processor, a memory storage device, and programs and instructions, accessible to the processor for executing the instructions utilizing the data stored in the memory storage device.

[0019] In other embodiments, the controller **122** may include manual controls that may be manipulated by an operator to control any of the procedures and equipment described herein. The controller **122** may also include a data acquisition system (DAS) **128**, which may include microprocessor-based components for collecting, displaying, recording, digitizing and storing data from BOP monitoring sensors **312** (FIG. 3). The DAS **128** may also supply the sensors **312** with an appropriate excitation voltage. The power system **124** may include generators for supplying electrical power to the wellbore system **100** as well as compressors, accumulators or other hydraulic equipment for supplying hydraulic power to the wellbore system **100**.

[0020] The BOP **102** may be coupled above the wellhead **116**, and may include one or more rams actuatable to seal around the conveyance **110**, or in some embodiments to sever the conveyance **110** and seal a passageway within the BOP **102**. Thus, the BOP **102** may prevent dangerous releases of gas or other wellbore fluids, which would otherwise endanger personnel or equipment at the surface location "S." The BOP **102** may be operably coupled to the DAS **128** by a communication cable **132** and operably coupled to the power system **124** by a hydraulic pressure supply line **134**.

[0021] Referring now to FIGS. 2A and 2B, the BOP **102** includes a central body **202** supported above the wellhead **116**. The central body **202** includes a central passageway **204** extending along a longitudinal axis X.sub.0. The central passageway **204** is fluidly coupled to the wellhead **116** and permits passage of the conveyance **110** therethrough. Extending radially from the body **202**, the BOP **102** includes a lower pair of opposed hydraulic cylinders **206a**, **206b** and an upper pair of opposed hydraulic cylinders **206c** and **206d** (generally or collectively, the hydraulic cylinders **206**). In other embodiments, more or fewer hydraulic cylinders **206** may be provided without departing from the scope of the disclosure. Each hydraulic cylinder **206** includes a piston **208** disposed therein with a ram **210** defined at an end thereof.

[0022] The pistons **208** are depicted in FIG. 2B an open configuration, where the rams **210** are retracted and otherwise disposed within the corresponding hydraulic cylinders **208**. The pistons **208** are selectively movable to a closed configuration, where the rams **210** are moved (advanced) into the body **202** to seal the central passageway **204**. Laterally opposing rams **210** may be shaped and otherwise configured to form a seal with one another **210** and/or with the conveyance **110** (FIG. 1)

such that flow through the central passageway **204** is prohibited when the rams **210** are moved to the closed configuration. In other embodiments, the rams **210** may include a sharpened edge and may thus be operable to sever the conveyance **110** before sealing the central passageway **204**. [0023] To move the pistons **208** to the closed configuration, a “close” chamber **212** within the hydraulic cylinders **206** may be pressurized to drive the pistons **208** and rams **210** radially inward (in the direction of arrow A.sub.1). To move the pistons **208** back to the open configuration, an “open” chamber **214** may be pressurized to drive the pistons **208** and rams **210** radially outward (in the direction of arrow A.sub.2).

[0024] The BOP **102** includes a control panel **220**. In the illustrated embodiment, the control panel **220** is disposed on an exterior of the body **202**. In other embodiments, the control panel **220** may be disposed in other locations including the controller **122**, DAS **128**, power system **124** and or other remote locations on or off the trailer **118** (FIG. 1). The control panel **202** may include manual or automated controls for pressurizing the “open” and “close” chambers, and for supporting other functionalities of the BOP **102**. The control panel **220** may provide connection points for the communication cable **132** and hydraulic pressure supply line **134** to operably couple the BOP **102** to the trailer **118** (FIG. 1) or other portions of a wellbore system.

[0025] Referring now to FIG. 3, a hydraulic circuit **300** is illustrated for controlling and monitoring each of the individual pistons **208** of the BOP **102**. The hydraulic circuit **300** receives a pressurized source of hydraulic fluid into the control panel **220** through the hydraulic pressure supply line **134**. In some embodiments, for example, the pressurized hydraulic fluid may include hydraulic oil pressurized to approximately 15,000 psi. The hydraulic pressure supply line **134** is fluidly coupled with a ram control valve **302** within the control panel **220**. The ram control valve **302** includes an open position **304** and a closed position **306** for appropriately directing the pressurized hydraulic fluid, and may be operated manually from the control panel **220** or alternatively, may be operated remotely from the controller **122** (FIG. 1) through the communication cable **132**. As illustrated in FIG. 3, the ram control valve **302** may comprise a selector type valve including a lever **307** that may be actuated to mechanically move the valve **302** between the open and closed positions, as recognized in the art.

[0026] A pair of hydraulic hoses **308**, **310** extend between the ram control valve **302** and an individual hydraulic cylinder **206b**. The first hydraulic hose **308** extends to the open chamber **214** and the second hydraulic hose **310** extends to the close chamber **212** such that the pressurized hydraulic fluid may be selectively delivered to the open and close chambers **214**, **212** to actuate the piston **208** as described above. The ram control valve **302** is illustrated with the lever **307** in a normally open position where the first hydraulic hose **308** is pressurized and the individual piston **208** may be maintained in the open or radially outward position described above. The lever **307** may be actuated to pressurize the second hydraulic hose **310** to move the piston **208** to the radially inward or closed position described above. Although only one hydraulic cylinder **206b** is illustrated in FIG. 3, it should be appreciated that each of the other hydraulic cylinders **206a**, **206c** and **206d** (FIG. 2B) may be similarly arranged with an individual hydraulic circuit for controlling the movement of a piston therein.

[0027] The hydraulic circuit **300** includes a plurality of monitoring sensors **312a**, **312b** **312c** (generally or collectively, monitoring sensors **312**) each communicatively coupled to the DAS **128** (FIG. 1) through the communication cable **132**. A panel monitoring sensor **312a** is coupled to the ram control valve **302** at the control panel **220**, and is operable to detect a parameter indicative of the real-time position of the ram control valve **302**. The first monitoring sensor **312a** may include proximity sensors, potentiometers, inclinometers (tilt sensors), or any other type of device operably coupled to any suitable component of the ram control valve **302** to provide a verification that the ram control valve **302** was moved to a position (e.g., an open or closed position) to which the ram control valve **302** and/or the lever **307** may have been instructed to move.

[0028] Individual circuit monitoring sensors **312b** are operatively coupled to each hydraulic hose

308, 310. In some embodiments, the second monitoring sensors **312b** may comprise pressure transducers that measure a pressure of the hydraulic fluid within the hydraulic hoses **308, 310** and/or turbine sensors (e.g., flow meters) that measure a flow rate and/or a direction of the hydraulic fluid within the hydraulic hoses **308, 310**. The second monitoring sensors **312b** may be employed to verify to that the hydraulic fluid moves through the hydraulic hoses **308, 310** as intended. In some embodiments, the second monitoring sensors **312b** may comprise flapper sensors or ultrasonic sensors that may detect any activity (e.g., any changing parameter) within the respective hydraulic hose **308, 310** or elsewhere in the hydraulic circuit **300**.

[0029] Piston monitoring sensors **312c** are positioned generally at the hydraulic cylinder **206b** or at the body **202** of the BOP **102** (FIG. 2B). In some embodiments, the piston monitoring sensors **312c** may comprise movement sensors (e.g., accelerometers) that detect a change in position of the pistons **208** within the hydraulic cylinder **206b**, and in other embodiments, the piston monitoring sensors **312c** may comprise proximity sensors that detect an actual position of the piston **208** within the hydraulic cylinder **206b**. It should be appreciated that any of the types of sensors described herein may be employed at various positions of the hydraulic circuit **300** to determine whether each of the pistons **208** were actuated as intended, or whether a failure has occurred at the control panel **220** (e.g., a switch or actuator failure), within the hydraulic hoses **308, 310** (e.g., leakages) or within the hydraulic cylinder **206b**. In some embodiments, if a leakage within the hydraulic circuit **300** occurs, data from the circuit monitoring sensors **312b** may be examined to determine in exactly which branch of the hydraulic circuit (e.g., **134, 308, 310**) the leakage occurred, or whether the leakage occurred within a hydraulic circuit associated with a different piston **208**. Each of the monitoring sensors **312** may transmit the data collected through the communication cable **132** for further processing and storage by the DAS **128** (FIG. 1).

[0030] FIG. 4 is a graph depicting example pressure detected within a hydraulic circuit of a BOP during a test operation. More specifically, a pressure of a hydraulic fluid within an accumulator for supplying hydraulic fluid to a hydraulic circuit **300** (FIG. 3) is graphed along the vertical axis. A time elapsed after initiating the test operation is graphed along the horizontal axis.

[0031] At point **402** the test operation begins with a pressure within the accumulator of about zero. After pressurizing the hydraulic fluid (e.g., with the power system **124**), the pressure may be raised to about 3000 psi, as indicated at point **404**. At point **406**, operation of the power system **124** may be interrupted such that the accumulator is not further pressurized. At point **408**, a piston of a BOP is instructed to close. The pressure drops as hydraulic fluid is supplied from the accumulator to operate the piston. The pressure drops to point **410** when the movement of the piston is complete. Similar pressure drops are observed at point **412** where the piston is re-opened, and at point **414** where the piston is closed once again. After the test operation is complete, a final pressure may be measured at point **416**. Where the accumulator supports a single piston, the functioning of the piston may be verified by the curve **400**. Where a BOP with multiple pistons is operated however, additional monitoring sensors **412** may be employed to verify the operation of any individual piston.

[0032] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, for example, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “contains”, “containing”, “includes”, “including,” “comprises”, and/or “comprising,” and variations thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0033] Terms of orientation are used herein merely for purposes of convention and referencing and are not to be construed as limiting. However, it is recognized these terms could be used with reference to an operator or user. Accordingly, no limitations are implied or to be inferred. In

addition, the use of ordinal numbers (e.g., first, second, third, etc.) is for distinction and not counting. For example, the use of “third” does not imply there must be a corresponding “first” or “second.” Also, if used herein, the terms “coupled” or “coupled to” or “connected” or “connected to” or “attached” or “attached to” may indicate establishing either a direct or indirect connection, and is not limited to either unless expressly referenced as such.

[0034] While the disclosure has described several exemplary embodiments, it will be understood by those skilled in the art that various changes can be made, and equivalents can be substituted for elements thereof, without departing from the spirit and scope of the invention. In addition, many modifications will be appreciated by those skilled in the art to adapt a particular instrument, situation, or material to embodiments of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiments disclosed, or to the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. Moreover, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, or component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

Claims

1. A rig-less wellbore system, comprising: a blowout preventer (BOP) providing a central body that defines a central passageway extending along a longitudinal axis through the central body, the central passageway fluidly coupled to a wellbore; a plurality of hydraulic cylinders extending radially from the central body; a plurality of pistons operable to move radially within a respective hydraulic cylinder between an open position, where flow through the central passageway is permitted, and a closed position, where the piston extends into the central passageway to prohibit fluid flow through the central passageway; at least one sensor operable to monitor at least one parameter indicative of activity within a hydraulic circuit operably associated with one of the pistons of the plurality of pistons; and a data acquisition system (DAS) communicatively coupled to the at least one sensor, the DAS operable to collect and record data provided by the at least one sensor.

2. The wellbore system of claim 1, further comprising a control panel and a ram control valve within the control panel, wherein the ram control valve is operable to move the one of the pistons between the open and closed positions.

3. The wellbore system of claim 2, wherein the at least one sensor comprises a panel sensor operable to detect a parameter indicative of a position of the ram control valve.

4. The wellbore system of claim 2, further comprising a pair of hydraulic hoses extending between the ram control valve and open and close chambers defined within the hydraulic cylinder in which the one of the pistons is disposed, wherein the one of the pistons is movable to the open position in response to pressurizing the open chamber and movable to the closed position in response to pressurizing the close chamber.

5. The wellbore system of claim 4, wherein the at least one sensor comprises a pressure sensor operable to monitor a pressure within one of the hydraulic hoses.

6. The wellbore system of claim 2, further comprising a source of pressurized hydraulic fluid disposed remotely with respect to the ram control valve and fluidly coupled to the control panel with a hydraulic fluid supply line.

7. The wellbore system of claim 6, wherein the DAS is disposed remotely with respect to the control panel and wherein the at least one sensor is communicatively coupled to the DAS through a communication cable extending between the DAS and the control panel.

8. The wellbore system of claim 1, wherein the at least one sensor is coupled to the hydraulic cylinder in which the one of the pistons is disposed and is operable to detect activity of the one of the pistons.

9. The wellbore system of claim 8, wherein the at least one sensor comprises a proximity sensor operable to detect an actual movement of the one of the pistons within the hydraulic cylinder.

10. The wellbore system of claim 1, further comprising a conveyance extending through the central body, wherein the conveyance includes one of a wireline, slickline, or coiled tubing.

11. A method of operating and monitoring a rig-less wellbore system, the method comprising: passing a conveyance through a BOP including a plurality of pistons; detecting, with at least one sensor, at least one parameter indicative of activity within a hydraulic circuit operably associated with an individual piston of the plurality of pistons; transmitting data collected by the at least one sensor about the at least one parameter through a communication cable to a DAS; and observing, with the DAS, a change in the data over time indicative of activity within the hydraulic circuit.

12. The method of claim 11, wherein passing the conveyance through the BOP includes passing one of a wireline, slickline, or coiled tubing through a central body of the BOP disposed radially inward of the plurality of pistons.

13. The method of claim 11, wherein detecting the at least one parameter includes detecting a pressure, a temperature, or a flow of a hydraulic fluid within a hydraulic hose extending between an open or close chamber of a hydraulic cylinder around the individual piston and a ram control valve.

14. The method of claim 13, further comprising instructing the ram control valve to move between an open position and a close position, and wherein detecting the at least one parameter further includes detecting the position of the ram control valve prior to and subsequent to instructing the ram control valve to move.

15. The method of claim 14, wherein detecting the at least one parameter further includes detecting a change in position of the individual piston within the hydraulic cylinder.
